

BEE 4750 Homework 1: Introduction to Using Julia

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Due Date

Thursday, 9/11/25, 9:00pm

Overview

Instructions

- Problem 1 consist of a series of code snippets for you to interpret and debug. You will be asked to identify relevant error(s) and fix the code.
- Problem 2 asks you to write code to implement several simple functions using more general programming and some which are unique to Julia (which might mean that you need to look at the documentation or use Julia's help function, which is accessed with `?`, e.g. `?sum` for help with the `sum()` function) or to analyze Julia syntax.
- Problem 3 asks you to convert a verbal description of a wastewater treatment system into a Julia function, and then to use that function to explore the impact of different wastewater allocation strategies.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `c:\Users\maozb\Documents\Cornell\BEE4750\Hw1\bee4750-hw1-maoz`

Standard Julia practice is to load all needed packages at the top of a file. If you need to load any additional packages in any assignments beyond those which are loaded by default, feel free to add a `using` statement, though [you may need to install the package](#).

```
using Random
using Plots
using GraphRecipes
using LaTeXStrings
using Distributions
```

```
# this sets a random seed, which ensures reproducibility of random number generation. You should
Random.seed!(1)
```

TaskLocalRNG()

Problems (Total: 30 Points)

Problem 1 (12 points)

The following subproblems all involve code snippets that require debugging. You are encouraged to use online resources (*e.g.* Julia documentation and forums, Stack Overflow, Reddit, etc) to help find diagnose error messages and find solutions, but make sure that you clearly document which resources you used and how you used them.

Problem 1.1

You've been tasked with writing code to identify the minimum value in an array. You cannot use a predefined function. Your colleague suggested the function below, but it does not return the minimum value.

```
function minimum(array)
    # initialize the minimum value counter
    min_value = array[1] # Problem 1.1: updated this line
    # update minimum values
    for i in 1:length(array)
        if array[i] < min_value
            min_value = array[i]
        end
    end
end
```

```

    end
    # return found minimum
    return min_value
end

array_values = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
@show minimum(array_values);

```

```
minimum(array_values) = 78
```

1.1 Write-up:

The logical flaw in the code was that the initial minimum value was set to zero as a placeholder, and the loop does not correct for this placeholder. Therefore, if the array does not contain any negative numbers, then the loop will not update the placeholder zero with an actual value from the array, and will return the placeholder zero as the minimum value. I solved this problem by using the first value in the array as a placeholder for the minimum value. This way, the loop will iterate through the array and only use values that are actually included in the array to determine the minimum value.

What is the logical flaw in the code? Fix it and show that your fixed code solves the problem.

Problem 1.2

Your team is trying to compute the average grade for your class, but the following code produces an `UndefVarError`.

```

# enter student grade vector
student_grades = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
# compute class average
function class_average(grades)
    average_grade = mean(grades) # Problem 1.2: changed this line
    return average_grade
end

average_grade = class_average(student_grades) # Problem 1.2: added this line
@show average_grade;

```

```
average_grade = 94.4
```

1.2 Write-up:

UndefVarError (undefined variable error) means that the variable `average_grade` is not defined. The code snippet returns an UndefVarError because `average_grade` is a variable defined inside the `class_average` function, but it is not defined as a variable outside of the function. The variable only exists inside the function but is not accessible directly without running the function.

Furthermore, inside the `class_average` function, the `mean` function calls the variable `student_grades`, which does not actually exist inside the function. It should instead take the mean of the “grades” input to the function.

To solve this error, I changed the code so that I am setting a variable `average_grade` equal to the output of the `class_average` function, so that this variable is actually defined before showing it. I also changed the `average_grade` line inside of the `class_average` function to take “grades” as an input since this is included in the function call.

What does this UndefVarError mean? Why does this code produce one? Fix the error and solve the problem.

Problem 1.3

Your team wants to know the expected payout of an old Italian dice game called *passadieci* (which was analyzed by Galileo as one of the first examples of a rigorous study of probability). The goal of *passadieci* is to get at least an 11 from rolling three fair, six-sided dice. Your strategy is to compute the average wins from 1,000 trials, but the code you’ve written below produces a `MethodError`.

```
# function to simulate a passedieci roll: 3 6-sided dice
function passedieci()
    # this rand() call samples 3 values from the vector [1, 6]
    roll = rand(1:6, 3)
    return roll
end
# set number of trials and initialize outcome vector
n_trials = 1_000
outcomes = zeros(n_trials)
# simulate number of passedieci rolls and count wins
for i = 1:n_trials
    outcomes[i] = (sum(passadieci()) > 11)
end
win_prob = sum(outcomes) / n_trials # compute average number of wins
@show win_prob;
```

```
win_prob = 0.374
```

1.3 Write-up:

The error is that “outcomes = zero(n_trials)” sets the variable outcomes equal to zero as an Int64 type, not as an array. Therefore, when the loop tries to set outcomes[i] equal to something, there results a `MethodError` because there is no setindex method for an integer since there is no index in an integer.

Changing the line to “outcomes = zeros(n_trials)” sets outcomes equal to an array of length n_trials, which makes it an iterable type which will have a setindex method.

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 1.4

You’re interested in writing some code to remove the mean of a vector from all of its components. You’ve written the following code and tried to test it on a random vector, but your code returns a `MethodError`.

```
# function to remove mean from a vector
function remove_mean(vect)
    # fucntion to compute the mean
    function compute_mean(vect)
        element_sum = 0 # initialize sum
        # compute mean and return
        for v in vect
            element_sum += v
        end
        return element_sum / length(vect)
    end

    m = compute_mean(vect) # compute mean
    # return demeaned vector
    vect .-= m
    return vect
end

random_vect = rand(1_000)
@show remove_mean(random_vect)
```

```
remove_mean(random_vect) = [0.3530897588497647, -0.22233920762541215, -0.4285281431874265, 0
```

```
1000-element Vector{Float64}:
```

```
 0.3530897588497647  
-0.22233920762541215  
-0.4285281431874265  
 0.042094947473510835  
-0.37439599509133203  
 0.479195069788551  
 0.3910893394555748  
-0.4471088588956105  
-0.029750755895614378  
-0.07344137600461798
```

```
 0.022854558539040504  
 0.22051538895596146  
-0.3971554491048144  
-0.04407972048258413  
 0.19308111762076452  
-0.25416286330908266  
-0.24374977941622034  
 0.3156914500557316  
 0.39261319239309367
```

1.4 Write-up:

The `MethodError` in this case comes from attempting to subtract an integer from a vector (array) type object, which is not possible since they do not have the same number of items within them.

The solution to this error is to use “`vect .-= m`” which individually subtracts `m` from each element inside of the variable `vect`, instead of just attempting to subtract an integer from the entire vector as a whole.

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Problem 2 (18 points)

Cheap Plastic Products, Inc. is operating a plant that produces $100\text{m}^3/\text{day}$ of wastewater that is discharged into Pristine Brook. The wastewater contains $1\text{kg}/\text{m}^3$ of YUK, a toxic substance. The US Environmental Protection Agency has imposed an effluent standard on the plant prohibiting discharge of more than $20\text{kg}/\text{day}$ of YUK into Pristine Brook.

Cheap Plastic Products has analyzed two methods for reducing its discharges of YUK. Method 1 is land disposal, which costs $X_1^2/20$ dollars per day, where X_1 is the amount of wastewater

disposed of on the land (m^3/day). With this method, 20% of the YUK applied to the land will eventually drain into the stream (*i.e.*, 80% of the YUK is removed by the soil).

Method 2 is a chemical treatment procedure which costs \$1.50 per m^3 of wastewater treated. The chemical treatment has an efficiency of $e = 1 - 0.005X_2$, where X_2 is the quantity of wastewater (m^3/day) treated. For example, if $X_2 = 50\text{m}^3/\text{day}$, then $e = 1 - 0.005(50) = 0.75$, so that 75% of the YUK is removed.

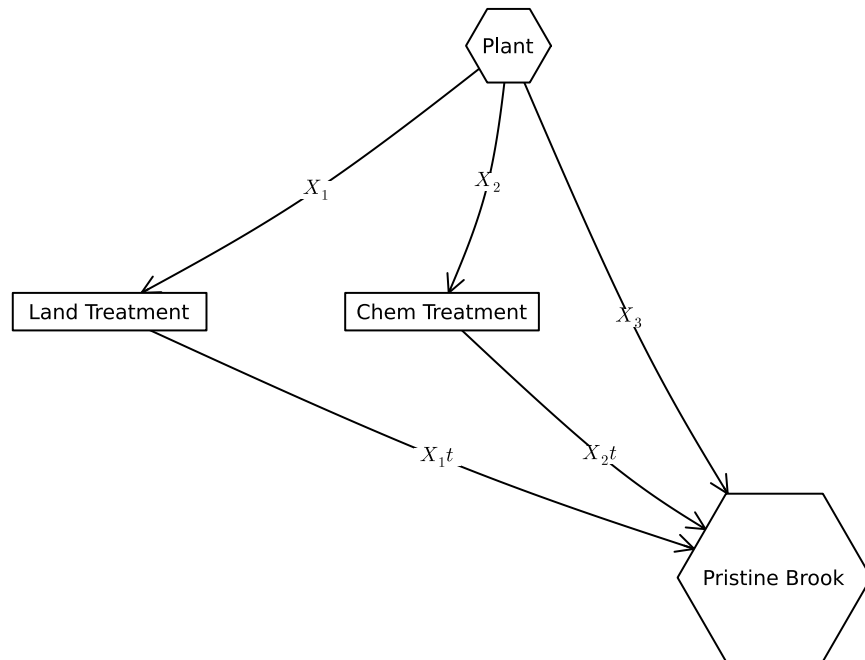
Cheap Plastic Products is wondering how to allocate their wastewater between these three disposal and treatment methods (land disposal, chemical treatment, and direct disposal) to meet the effluent standard while keeping costs manageable.

The flow of wastewater through this treatment system is shown in Figure 1. Modify the edge labels (by editing the `edge_labels` dictionary in the code producing Figure 1) to show how the wastewater allocations result in the final YUK discharge into Pristine Brook. For the `edge_label` dictionary, the tuple (i, j) corresponds to the arrow going from node i to node j . The syntax for any entry is $(i, j) \Rightarrow \text{"label text"}$, and the label text can include mathematical notation if the string is prefaced with an L, as in `L"x_1"` will produce x_1 .

```
A = [0 1 1 1;
      0 0 0 1;
      0 0 0 1;
      0 0 0 0]

names = ["Plant", "Land Treatment", "Chem Treatment", "Pristine Brook"]
# modify this dictionary to add labels
edge_labels = Dict((1, 2) => L"x_1", (1,3) => L"x_2", (1, 4) => L"x_3", (2, 4) => L"x_1t", (3,
shapes=[:hexagon, :rect, :rect, :hexagon]
xpos = [0, -1.5, -0.25, 1]
ypos = [1, 0, 0, -1]

p = graphplot(A, names=names, edgelabel=edge_labels, markersize=0.15, markershapes=shapes, ma
display(p)
```



Problem 2.1

Formulate a mathematical model for the treatment cost and the amount of YUK that will be discharged into Pristine Brook based on the wastewater allocations. This is best done with some equations and supporting text explaining the derivation. Make sure you include, as additional equations in the model, any needed constraints on relevant values. You can find some basics on writing mathematical equations using the LaTeX typesetting syntax [here](#), and a cheatsheet with LaTeX commands can be found on the course website's [Resources page](#).

2.1 Equations:

$$X_1 + X_2 + X_3 = 100 \text{ (Plant produces } 100 \text{ m}^3 \text{ of wastewater)}$$

$$X_1t = 0.2 * X_1 \text{ (Output of land treatment)}$$

$$X_2t = (0.005 * X_2) * X_2 \text{ (Output of chemical treatment)}$$

$$W_d = X_1t + X_2t + X_3 \text{ (Wastewater discharged into Pristine Brook)}$$

$$C = X_1^2/20 + 1.50 * X_2 \text{ (Combined cost of treatment options)}$$

Problem 2.2

Implement your systems model as a Julia function which computes the resulting YUK discharge and cost for a particular treatment plan. You can return multiple values from a function with a [tuple](#), as in:

```
function multiple_return_values(x, y)
    return (x+y, x*y)
end

function yuk_discharge(x1, x2) # function for YUK discharge, x1 = diversion to land treatment
    x3 = 100 - x1 - x2 # Plant produces 100 m^3 of wastewater
    x1t = 0.2 * x1 # Output of land treatment
    x2t = 0.005 * x2 * x2 # Output of chemical treatment
    wd = x1t + x2t + x3 # Wastewater discharged into Pristine Brook
    C = x1 * x1 / 20 + 1.5 * x2 # Combined cost of treatment options
    return(wd, C) # Returns wastewater discharge into the brook and cost
end

wd, C = yuk_discharge(10, 90)
@show wd;
@show C;
```

```
wd = 42.5
C = 140.0
```

To evaluate the function over vectors of inputs, you can *broadcast* the function by adding a decimal `.` before the function arguments and accessing the resulting values by writing a *comprehension* to loop over the individual outputs in the vector:

```
x = [1, 2, 3, 4, 5]
y = [6, 7, 8, 9, 10]

output = yuk_discharge.(x, y) # Changed to new YUK discharge function
wd = [out[1] for out in output] # Changed variable name to wd (wastewater discharged to brook)
c = [out[2] for out in output] # Changed variable name to c (cost)
@show wd;
@show c;
```

```
wd = [93.38, 91.645, 89.92, 88.205, 86.5]
c = [9.05, 10.7, 12.45, 14.3, 16.25]
```

2.3 Write-up:

I added a new function to use the equations from problem 2.1. The function takes waste diversion to land and chemical treatment options as input, and assumes the rest is untreated. The function uses the equations to solve for amount of polluted wastewater discharged to the Pristine Brook, and the cost of the given diversion plan.

Make sure you comment your code appropriately to make it clear what is going on and why.

Problem 2.3

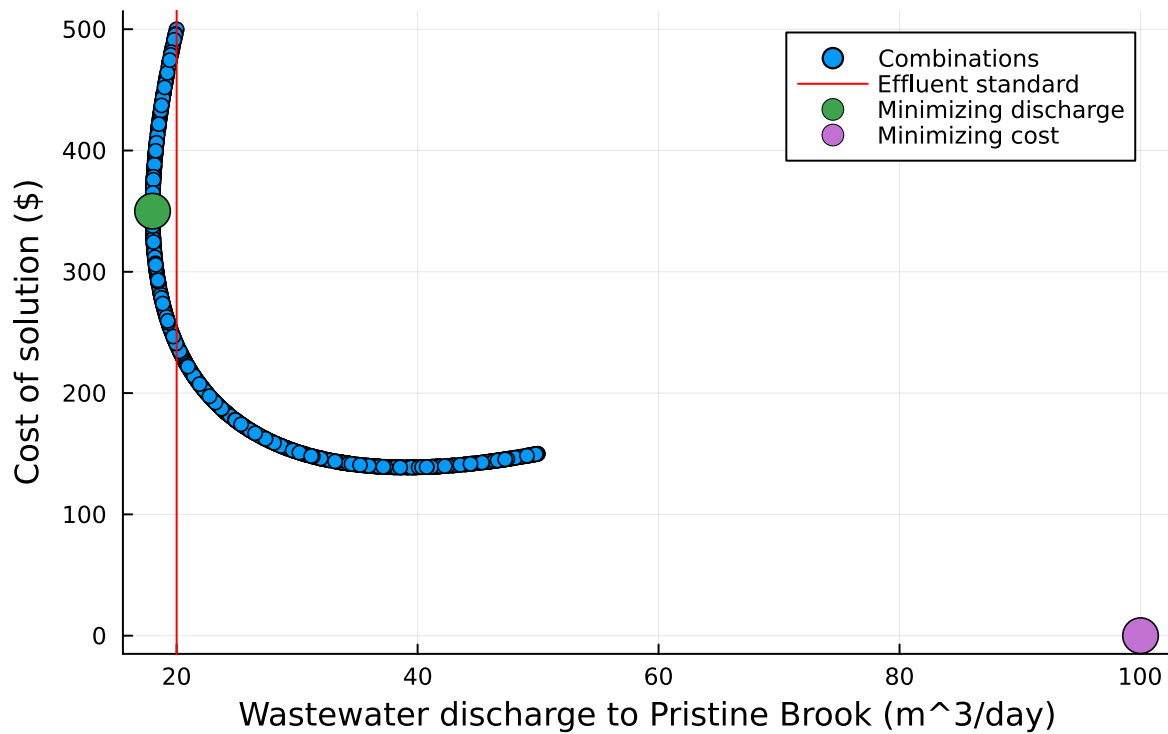
Use your function to experiment with 1,000 different combinations of wastewater discharge and treatment. You can do this with either a grid search or by sampling from a [Dirichlet distribution](#) (a $\text{Dirichlet}(1, n)$ distribution will generate uniformly-weighted n -dimensional vectors whose components add up to 1; see the [Distributions.jl documentation](#) for how to sample from probability distributions in Julia). Plot the results of these experiments. Do any satisfy the YUK effluent standard (plot this as well as a dashed red line). What was the cost of solutions satisfying the standard? What can you say about the tradeoff between treatment cost and YUK discharge? You don't have to find an "optimal" solution to this problem, but what do you think would be needed to find a better solution?

```
# Making the 1000 combinations with a Dirichlet distribution
d = Dirichlet([1,1])
samples = rand(d,1000) .* 100
pairs = permutedims(samples)

# Running YUK discharge function
results = [yuk_discharge(pairs[i,1], pairs[i,2]) for i in 1:size(pairs,1)]
wds = [r[1] for r in results]
cs = [r[2] for r in results]

# Plotting results
p = scatter(wds,cs,xlabel="Wastewater discharge to Pristine Brook (m^3/day)",ylabel="Cost of
vline!([20],color=:red,label="Effluent standard")
scatter!([18],[350],label="Minimizing discharge",markersize=10)
scatter!([100],[0],label="Minimizing cost",markersize=10)
display(p)
```

YUK Treatment Options



2.3 Write-up:

Yes, many of the solutions satisfy the YUK effluent standard! The minimum cost for a solution that satisfies this standard is around \$240, which is not the cheapest but still relatively low compared to the high price of other solutions. There appears to be a tradeoff, where more effective solutions cost more, but there are also some relatively affordable solutions that do meet the standard.

I think that to find an optimal or better solution, there should be a “penalty” for any amount of wastewater discharge, since discharge would likely incur some sort of costs, financial, environmental, or otherwise. This would allow for a more robust model which would find a single optimal solution. In this case, however, I think that the optimal solution would be the cheapest one (~\$240) which falls below the effluent standard.

Problem 2.4

Find the strategies which minimize cost and YUK discharge (these will be different strategies) analytically and find the values of the objective metrics. Plot these values in the plot that you created for Problem 2.3. How do their values compare to the spread of values that you found in that problem? Would you select either of them (explain why or why not)?

The strategy to minimize cost would be to minimize the equation $C = X_1^2/20 + 1.50 * X_2$ (Combined cost of treatment options). This would result in an X_1 and an X_2 of zero, since discharging waste into the Brook is free.

The strategy to minimize YUK discharge would be to minimize the equations:

$$X_1 + X_2 + X_3 = 100 \text{ (Plant produces } 100 \text{ m}^3 \text{ of wastewater)}$$

$$X_1 t = 0.2 * X_1 \text{ (Output of land treatment)}$$

$$X_2 t = (0.005 * X_2) * X_2 \text{ (Output of chemical treatment)}$$

$$W_d = X_1 t + X_2 t + X_3 \text{ (Wastewater discharged into Pristine Brook)}$$

This can be simplified:

$$W_d = 0.2 * X_1 + 0.005 * X_2^2 + (100 - X_1 - X_2) = 100 - 0.8 * X_1 - X_2 + 0.005 * X_2^2$$

Minimizing the equation, $W_d = 18m^3/d$.

At this point, $X_1 = 80$ and $X_2 = 20$, and $X_3 = 0$ which was possible to predict beforehand since X_3 will always release more waste than X_1 .

Therefore, cost is:

$$C = X_1^2/20 + 1.50 * X_2 = (80^2)/20 + 1.50 * (20) = 350.$$

This represents the left-most point on my graph, where waste is at a minimum but cost is somewhat high.

Personally, I would select the option around \$240, since this is the cheapest way to avoid exceeding the YUK standard, assuming that there is no other incentive to further reduce discharge amount.

References

List any external references consulted, including classmates.